

Question	Marking guidance			Mark	Comments
07.1	1. Macerate / homogenise / blend / break tissues / cells (in solution); 2. Centrifuge; 3. At different / increasing speeds until chloroplast fraction obtained;			3	1. Accept any suitable method to break tissues / cells / release organelles 2. and 3. Allow 'perform differential centrifugation until chloroplasts obtained.' for 2 marks
07.2	Feature	Mitochondrion	Chloroplast	3	1 mark for each correct row Crosses = blank space
	Double outer membrane	✓	✓		
	Starch grains		✓		
	Diffusion of oxygen into the organelle	✓			
07.3	The site of aerobic respiration (reactions) OR ATP is made / ADP is phosphorylated;			1	Reject 'energy is produced'
07.4	1. Training made no difference to number (of mitochondria per μm^2); 2. Training led to an increase in the area (of inner mitochondrial membrane);			2	1. Accept Figure 7 as mean number of mitochondria per μm^2 2. Accept Figure 8 as area of inner mitochondrial membrane